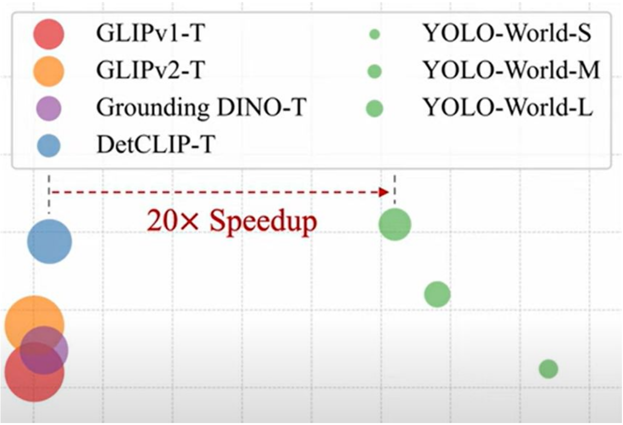
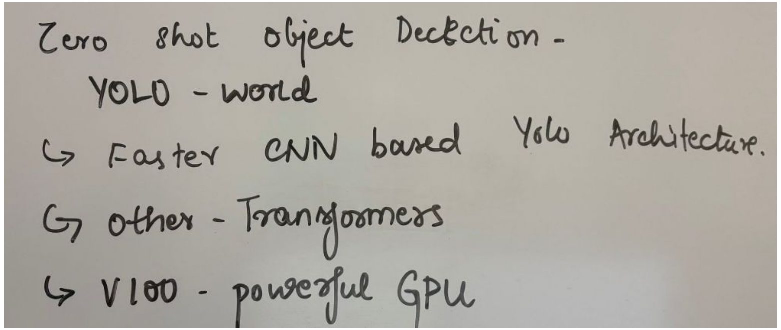
Object detection

Real time, zero shot object detection with YOLO World

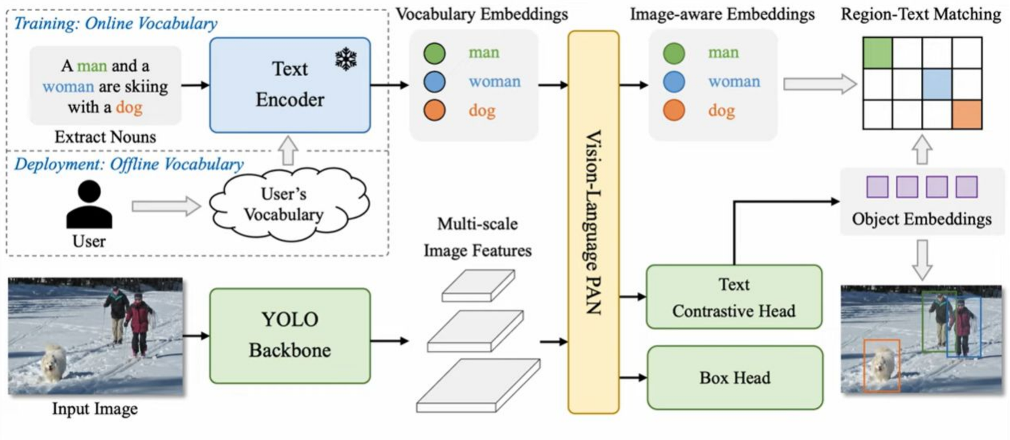
Introduction

- Yolo world - zero short 20 times faster than its precede
- Traditional - pre designed and only for number of set of objects
- Open vocabulary model - like grounding DINO (zero shot) - prompting the model on what we are looking for - speed is more





Model Architecture



- Yolo world has 3 key parts.
- Yolo detector - extracts multi scale features from input image clip.
- Clip text encoder encodes the text into text embeddings.
- Custom Networks that performs multi level; cross modality fusion between image features and text embeddings.
- Embeddings - passed and reused.

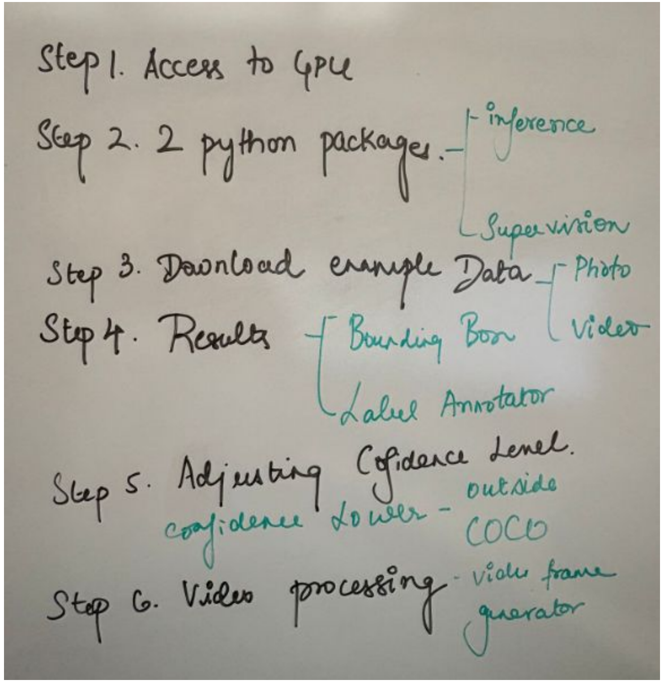
Links to notebooks

Zero short object detection with YOLO world

[zero-shot-object-detection-with-yolo-world.ipynb - Colab]()

Ensure :

Colab is GPU accelerated



Explanation

- Roboflow - python package
- Import the packages needed
- YOLO world - 4 different sizes
- YOLO world classes - we can choose - clip model is downloaded in the background
- From class only it will be detected
- The classes outside COCO dataset will be less confidence
- More classes detected - problem - duplication - non max suppression - use intersection over union - IOU - overlap issue
- With nms function - next

Next Experiment - yellow substance

- Locating the filters
- Choosing the right prompt - is necessary
- Hard to define
- Color reference - effective
- Set classes method
- Detects individual holes - entire object (challenge)
- High level bounding boxes
- Filter - based on relative area
- Calculating total area with pixels

Continued…

- Result - only filled holes
- Process entire video and save it - recorded
- All results goes inside the for loop
- YOLO world - golden solution
- Latency - much faster than predecessor , small than real time object detector